

Diode Model :

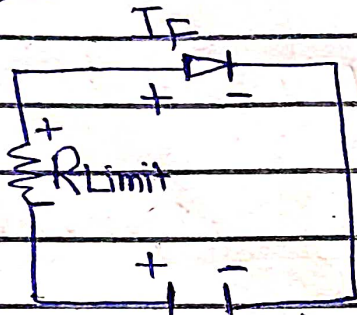
Ideal
Diode
Model

Practical
diode
Model

Complete
diode
Model

① - Ideal Diode Model:

Forward Bias :

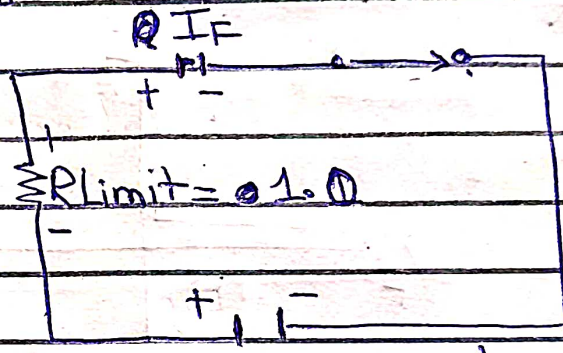


V_{Bais}

Simple
diagram

① $I_{in} = 0$

③ $V_F = ?$



$V_{Bais} = 10$

Ideal
Diagram

② $I_F = ?$

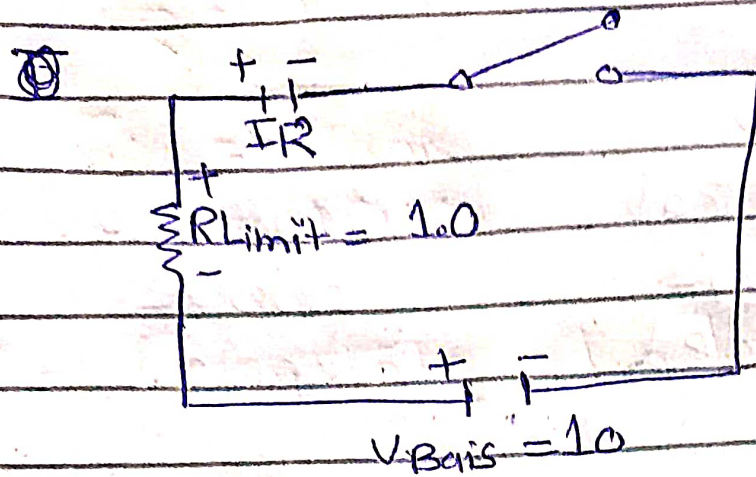
$$R_{Limit} + I_F \cdot 0 = V_{Bais}$$

$$I_F = \frac{V_{Bais}}{R_{Limit}}$$

$$I_F = \frac{10}{1.0} = 10$$

$$V_F = I_F \times R_{Limit} = 10 \times 1.0 = 10$$

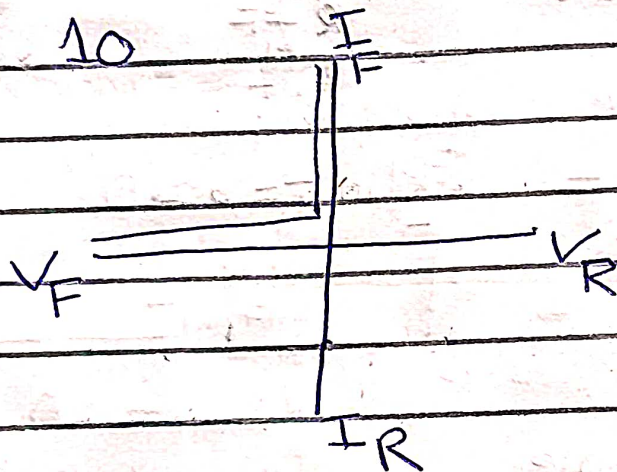
Reverse Bias:



$$I_{in} = 0$$

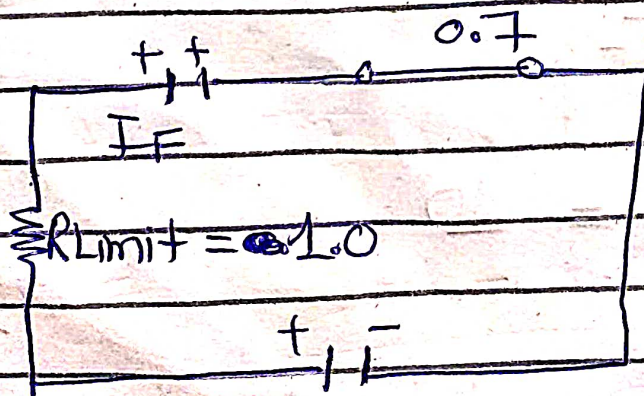
$$I_R = 0$$

$$V_R = V_{Bias} = 10$$



② - Practical Model:

forward Bias:



$$I_{in} = 0$$

~~$$I_{p_acc}$$~~
$$I_F$$

$$R_{limit} + I_F + 0.7 + V_{Bias} = 0$$

$$I_F = \frac{0.7 + V_{Bias}}{R_{limit}}$$

$$I_F = \frac{0.7 + 10}{10}$$

$$I_F = 9.3 \text{ A}$$

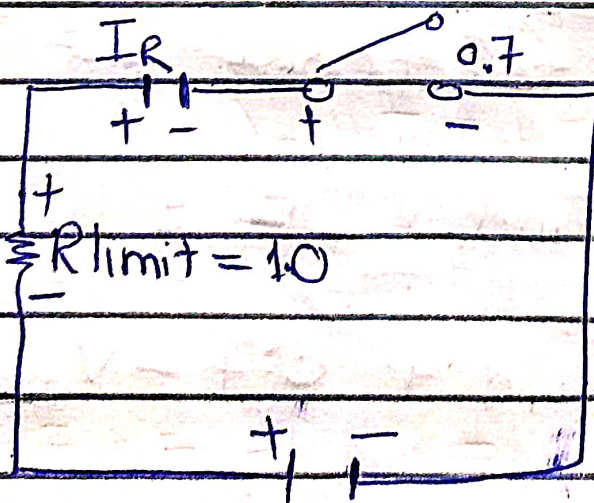
$$V_F = I_F \times R_{limit}$$

~~9.3 x 1.0~~

$$9.3 \times 1.0$$

$$= 9.3 \text{ V}$$

Reverse Bias



$$V_{Bias} = 10$$

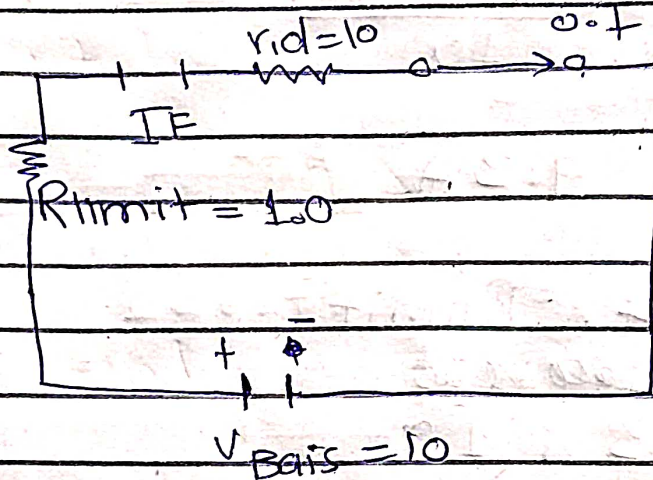
$$I_{in} = 0$$

$$I_R = 0$$

$$V_R = V_{Bias} = 10V$$

③ - Complete Diode Model:

Forward Bias:



$$I_{in} = r_{d} = 10$$

~~$I_F = 0$~~

~~$I_F = 0$~~ Now Find I_F :

$$R_{limit} + I_F + r_d + 0.7 - V_{Bias}$$

$$I_F = \frac{V_{Bias} - 0.7}{r_d + R_{limit}}$$

$$I_F = \frac{10 - 0.7}{10 + 1.0}$$

$$= \frac{10 - 0.7}{10 + 1.0}$$

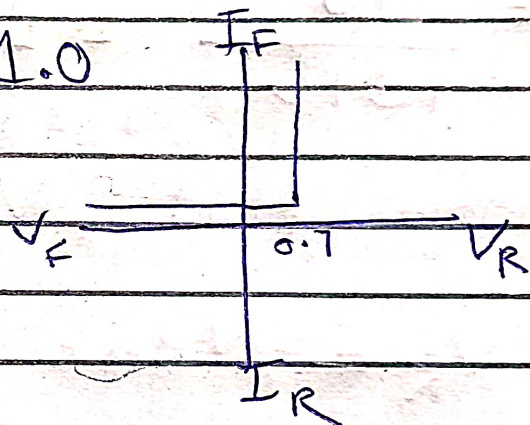
$$= \frac{9.3}{10.0} = 0.93 \text{ A}$$

Now find $V_F = ?$

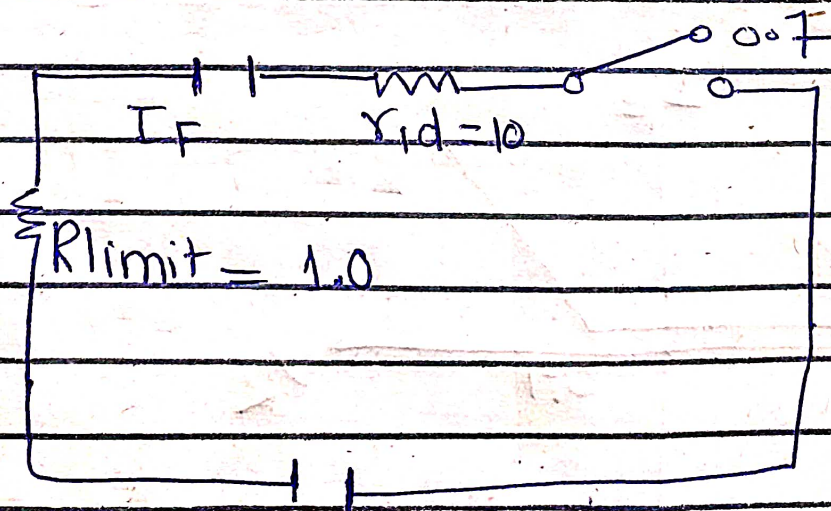
$$V_F = I_F \times R_{\text{limit}}$$

$$V_F = 0.93 \times 1.0$$

$$V_F = 0.93 \text{ V}$$



Reverse Bias :-



$$V_{\text{Bias}} = 10$$

$$\textcircled{1} \quad I_{in} = r_{id}$$

$$= 10$$

$$\textcircled{2} \quad I_R =$$

$$= \frac{V_{Bias}}{r_{id} + R_{limit}}$$

$$= \frac{10}{10 + 1.0}$$

$$= \frac{10}{10.10} = 1A$$

now find $V_F = ?$

$$V_R = I_R V_{Bias}$$

$$~~V_R~~ \quad V_R = 1 \times 10 = 10V$$

